

27	A sinusoidal current of peak value I_0 is passed through a resistor R with a diode connected in series with it.							
	What is the average rate of heat dissipated in R ?							
	A	$0.25 I_0^2 R$	B	$0.50 I_0^2 R$	C	$0.71 I_0^2 R$	D	$I_0^2 R$
	Answer: A							
	For half wave rectification, $I_{\text{r.m.s.}} = \frac{I_0}{2}$							
	Power = $I_{\text{r.m.s.}}^2 R$ $= 0.25 I_0^2 R$							

28	The de Broglie wavelength of a particle that has kinetic energy X is ?
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